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Beyond Opportunistic Selection: A Customizable, Multi-Objective Framework for Country-Scale CMIP6 Sub-Ensembles

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The growing volume and structural diversity of the CMIP6 archive [1] has made the selection of representative models for regional and country-scale impact assessments increasingly non-trivial. Although full-ensemble approaches are valuable for characterizing uncertainty, computational and operational constraints often require downstream users, for example dynamical downscaling initiatives such as CORDEX, sectoral impact modelling, and climate risk assessment, to work with small sub-ensembles. These subsets are commonly chosen opportunistically or inherited from static lists, which can under-sample plausible regional futures and over-represent closely related models. We present a configurable framework for selecting regionally tailored sub-ensembles from CMIP6 when computational or operational constraints preclude using the full ensemble. The framework integrates three decision dimensions: model independence, historical fidelity, and representativeness of the projected response spread of the full CMIP6 ensemble.

Model independence is quantified via unsupervised learning by embedding models in a feature space derived from regional climate responses and clustering them into families, enabling the selection procedure to reduce redundancy by avoiding highly similar model behaviour. Historical fidelity is assessed using core variables (near-surface air temperature, precipitation, and sea-level pressure) and complementary metrics that summarize both bias magnitude and pattern fidelity. These are combined into a composite score that penalizes single-metric failure while remaining interpretable. To preserve coverage of plausible regional futures, models are simultaneously evaluated in a future-response space defined by end-of-century changes in temperature and precipitation, with the option to include additional proxies relevant to extremes. In the spirit of recent independence–performance–spread selection approaches (e.g., ClimSIPS [2]), we emphasize country-scale customization, explicit trade-offs, and fully transparent diagnostics.

The criteria are integrated into a multi-objective selection engine that recommends subsets of a user-specified size. The process is customizable, allowing users to adjust weights assigned to performance, spread coverage, and independence, and to impose constraints on spatial resolution. We illustrate how recommended subsets can differ across contrasting climatic regions and user priorities, supporting robust and documented model selection for regional assessments and downscaling workflows.

References

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